Why

The edge of a wildfire is the most dangerous place for a firefighter.

Robot dogs (four-legged robots) can climb steep, rocky, smoky ground where trucks get stuck and drones can't see.

> **But a robot near a fire must know where the fire *will* be, not just where it is now.**

How we predict the fire

Real fire history
~246,000 NASA satellite fire detections from 30 major wildfires worldwide

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Weather
wind, humidity, temperature, rain

Terrain
elevation and slope

Land cover
forest, shrub, grass, urban

Fire shape
nearby burning cells, fire size

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Machine learning: XGBoost
27 inputs per 50 m cell. Trained on 24 fires, tested on 6 fires it never saw.

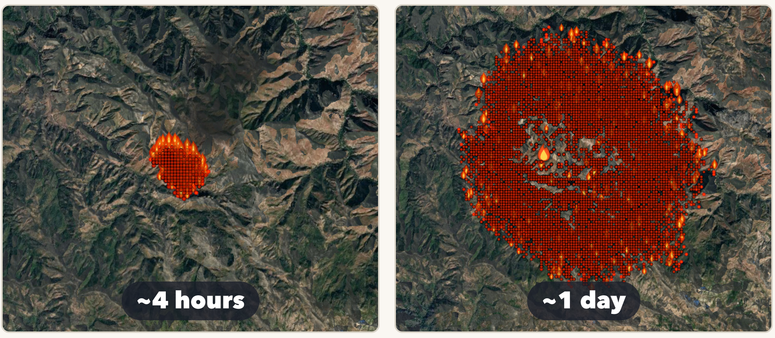
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Physics: Rothermel spread equation
fuel × wind × slope × dryness. The ML model adjusts the rate.

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Output
the chance each 50 m cell catches fire, every time step

The simulator



Simulated fire spreading over real 3D terrain in the Big Sur area of Monterey County, California (summer weather). Runs in a web browser on Google 3D map tiles.

Results

Tested on **6 fires the model never saw** (USA, Bolivia, Siberia, Greece).

In these fires 91% of cases are "fire spreads," so we compare against a simple baseline that **always guesses "spreads."**

Ranking risky cells (AUC-ROC, higher is better)

Our model	**0.75**
Baseline	**0.50**

Probability error (Brier score, lower is better)

Our model	**0.076**
Baseline	**0.083**

Takeaway: the model ranks which cells are at risk clearly better than chance, and its probabilities are more accurate than the baseline. It still leans toward "spreads," which we are working to improve.